Project Report

HEALTHCARE ANALYTICS DASHBOARD

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Tools Used: Excel and Power BI

Data Science & Analytics

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Documentation

1. Data Model Design Decisions

The data model follows a star schema design. Admissions is the central fact table connected to dimension tables including Departments, Patients, Doctors, Diagnosis, and Date. Dimension tables are not directly related to each other to maintain model performance and accuracy.

2. Key DAX Formulas and Their Purpose

Total Admissions = COUNT(Admissions[AdmissionID]) → Counts total admissions.
Avg LOS = AVERAGE(Admissions[Length_Of_Stay]) → Calculates average length of stay.
Readmission Rate % = DIVIDE(CALCULATE(COUNT(Admissions[AdmissionID]), Admissions[ReadmissionFlag]="Yes"), COUNT(Admissions[AdmissionID])) → Measures readmission percentage.
Department Rank = RANKX(ALLSELECTED(Departments[DepartmentName]), [Total Admissions], , DESC) → Ranks departments by performance.
Capacity Utilization % = DIVIDE([Beds Used], [Bed Capacity Value]) → Shows hospital load.

3. Top 5 Insights from Analysis

- Cardiology and Oncology departments handle the highest patient volume.
- Oncology shows the longest average length of stay.
- Departments with high LOS tend to have higher treatment costs.
- General Medicine department has lowest cost per patient.

4. Conclusion

This dashboard provides a comprehensive decision-support system for hospital administrators. It enables performance monitoring, operational risk detection, and strategic planning through interactive visual analytics.